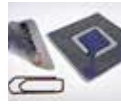




Twitter for Android TV

Twitter launches a new app for Android TV for live streams and support for Periscope
<http://dgit.in/twtrAtv>

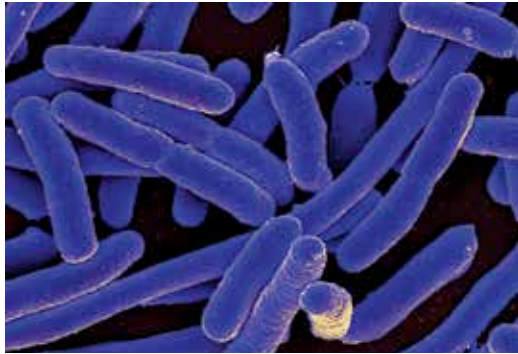


Heart monitor sticker

Scientists develop a new stick-on wearable sensor as tiny as a fingernail that can monitor heart rates. <http://dgit.in/stckrhrt>

From the labs

DNA by treating base pairs (A,C,T,G) as bit values (0,1). Since the advent of DNA synthesisers and sequencers, this is almost an ordinary feat, as the hardest part would be deciding the



Do you see E. Coli, or do you see nano pen drives?

most robust coding system to use to encode the information. The primary advantages of storing information in DNA are high density and longevity. One gram of DNA can hold upto 455 exabytes of data, according to researchers at the Swiss Federal Institute of Technology. Researchers at Harvard have demonstrated storage of 700 terabytes in a single gram; that's the equivalent of 14,000 50GB Blu-Ray discs! With DNA storage, a small vial would suffice for storing all the information that ever existed! At temperatures between 60 and 70 degrees Celsius, the information stored in DNA strands was readable for at least a week, so at much lower temperatures like at the Arctic Seed Vault, the information would last for millions of years! This storage however, though made of the stuff of life, is not strictly living. The recent breakthrough by Harvard geneticists has resulted in moving, squirming hard drives, as they managed to encode the information in the genome of living bacteria cells. To actually transfer the information, they used a natural immune system found in certain strains of bacteria called the CRISPR/Cas system. This biological functionality allows bacteria cells to protect themselves from invading viruses. They do this essentially by copy pasting certain parts of the viral DNA and assimilating it into a specific

location in their own genome to learn and remember the immunity to that particular virus. This genetic information is passed on as the bacteria live out their life cycles and so the information remains stored in the colony as it evolves. To read out the information, the researchers basically made their data containing DNA look like viral DNA by attaching similar ends to the DNA strands, and then introduced it in the bacteria culture. They did this in a colony of E. Coli bacteria that possessed the CRISPR/Cas system. An added benefit is that the bacteria stores the 'immunity information'

(as far as it is concerned) sequentially, and thus it makes it possible to have longer strings of concatenated data by introducing the strands in that order. In their experiment, the geneticists were able to store 100 bytes of data, which can easily be extended by finding more suitable strains of bacteria or modifying existing ones. However, there is also a slight imperfection in the process as there's no guarantee that all of the individual bacteria cells will take up the introduced 'viral' code. For example, if we are storing the sentence "DIGIT IS THE BEST TECH MAGAZINE", and we introduce the words one by one in individual strands, there is a chance that many individuals in the colony will have incomplete (but ordered) sentences like "IS BEST TECH MAGAZINE", "DIGIT IS BEST TECH", etc.

Even though it's not perfect, this can easily be overcome by sequencing enough individual cells. Perhaps one day all our medical information can be stored and passed down with our genes. Though we're not there yet, at least it's the first step!

Biots

A term coined by science fiction author Arthur C. Clarke, biological robots use living muscle tissue for

locomotion. Though today's biots can hardly crawl, inching forward at a slow pace in specialised conditions, they are surely inching forward. Calling them 'bots' may be an overstatement for now but we're getting there! The first time a bio bot was created, scientists composed tissue from heart cells. Later, researchers in the University of Illinois succeeded in fabricating a working bio bot from muscle cells. The process is basically composed of building a scaffolding and laying the tissue cells. This starts with 3D printing of a scaffolding made of hydrogel, which is a material that is similar to soft contact lenses. The scaffolding is then substantiated by depositing skeletal muscle cells of a rat, which connects the ends of the scaffolding. They did this to make 6mm long strips of bio-bots, which they placed in a petri dish with a warm conducting liquid. To make the bio bots move, they sent electrical pulses into the dish, causing the muscle tissue to expand and contract repeatedly,



Print your own biots at home

moving forward like a caterpillar. In the future, the movement can be augmented with nerve cells for sensing capabilities. They will also need a circulatory system or specialised exoskeleton if they are to survive outside a warm fluid and possibly be scaled to a usefully large size. If you like the sound of this and have ideas to test, you can even buy your own 3D printer for living tissue at biobots.io! 